

NAME:

NO:

CLASS:

RIVERSIDE SECONDARY SCHOOL



PRELIMINARY EXAMINATION 2026

SUBJECT : MATHEMATICS
SUBJECT CODE /PAPER : 4052/01
LEVEL/STREAM : Sec 4 Express, 5 Normal (Academic)
DURATION : 2 hours 15 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your class, index number and name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

If working is needed for any question it must be shown with the answer.

Omission of essential working will result in loss of marks.

The use of an approved scientific calculator is expected, where appropriate.

If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to 1 decimal place.

For π , use either your calculator value or 3.142, unless the question requires the answer in terms of π .

At the end of the examination fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

The total of the marks for this paper is 90.

Submit Part A and Part B separately.

Mathematical Formulae*Compound interest*

$$\text{Total amount} = P \left(1 + \frac{r}{100} \right)^n$$

Mensuration

$$\text{Curved surface area of a cone} = \pi r l$$

$$\text{Surface area of a sphere} = 4\pi r^2$$

$$\text{Volume of a cone} = \frac{1}{3} \pi r^2 h$$

$$\text{Volume of a sphere} = \frac{4}{3} \pi r^3$$

$$\text{Area of triangle } ABC = \frac{1}{2} ab \sin C$$

$$\text{Arc length} = r\theta, \text{ where } \theta \text{ is in radians}$$

$$\text{Sector area} = \frac{1}{2} r^2 \theta, \text{ where } \theta \text{ is in radians}$$

Trigonometry

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

Statistics

$$\text{Mean} = \frac{\sum fx}{\sum f}$$

$$\text{Standard deviation} = \sqrt{\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f} \right)^2}$$

Part AAnswer **all** the questions.

- 1 (a) Express 0.000156 in standard form.

Answer [1]

- (b) Work out $(8.3 \times 10^{150}) - (6.5 \times 10^{149})$.

Give your answer in standard form.

Answer [1]

-
- 2 The bearing of Q from P is 072° .

Find the bearing of P from Q .

Answer [2]

- 3 Given that $8^{y+1} = 32^3$, find y .

Answer $y = \dots\dots\dots$ [2]

- 4 $\sin\left(\frac{x}{2}\right)^\circ = 0.6$

Find two possible values of x in the range $0 \leq x \leq 360$.

Answer $x = \dots\dots\dots$ or $\dots\dots\dots$ [2]

- 5 Expand and simplify $(2x-3)(x+5)-(x-1)^2$.

Answer

Answer [3]

-
- 6 A collector has a collection of Pokémon cards.
34% of the cards are 'Common' rarity.
Of the remaining cards, 60% are 'Rare' rarity.
The other 132 cards are 'Legendary' rarity.
Calculate the total number of cards in the collection.

Answer cards [3]

7 The sum of the interior angles of a regular polygon is 2340° .

(a) Find the number of sides of the polygon.

Answer sides [2]

(b) Hence, find the size of each exterior angle of the polygon.

Answer [1]

8 $p = \frac{2x+8}{x+1}$

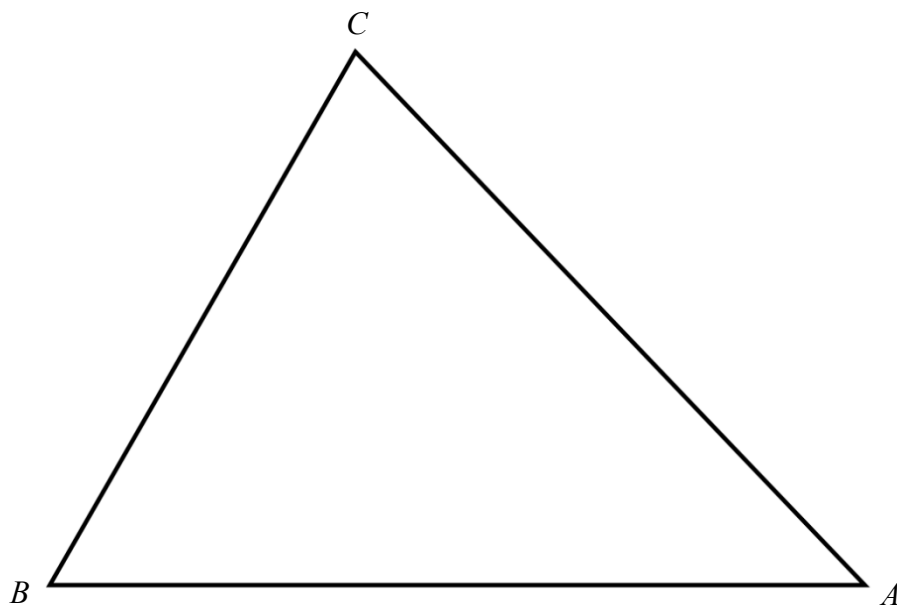
(a) Make x the subject of the formula.

Answer $x =$ [2]

(b) Hence, find the value of x when $p = 3$.

Answer $x =$ [1]

- 9 The diagram shows triangle ABC .



(a) Construct the perpendicular bisector of AB . [1]

(b) Construct the bisector of angle ABC . [1]

(c) A point P lies inside the triangle such that P is nearer to A than to B , and nearer to BA than to BC .

Shade the region in which P must lie. [1]

- 10 The intensity of light, I lux, received from a light source is inversely proportional to the square of the distance, d metres, from the source. The distance from the source is increased by 60% of its original value. Calculate the percentage reduction in the intensity of light.

Answer% [3]

- 11** The stem-and-leaf diagram shows the reaction times, in seconds, of 12 drivers when responding to an unexpected hazard during a driving assessment.

2		3	5	8			
3		1	2	5	<i>c</i>	8	9
4		1	4	7			

Key: 3 | 1 represents 3.1 seconds

The median of the data is 3.65.

Find

- (a)** the interquartile range of reaction times,

Answer seconds [1]

- (b)** the value of *c*.

Answer *c* = [2]

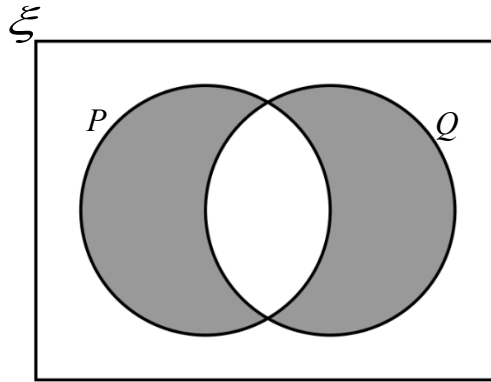
- 12** A company manufactures two geometrically similar replica of World Cup trophies: a standard replica and a large replica.
- The amount of gold plating needed for the large replica is 4 times the amount of gold plating needed for the standard replica.
- (a)** Find the ratio of the height of the standard replica to the height of the large replica.

Answer : [1]

- (b)** The standard replica has a mass of 300 g.
- Calculate the mass of the large replica.

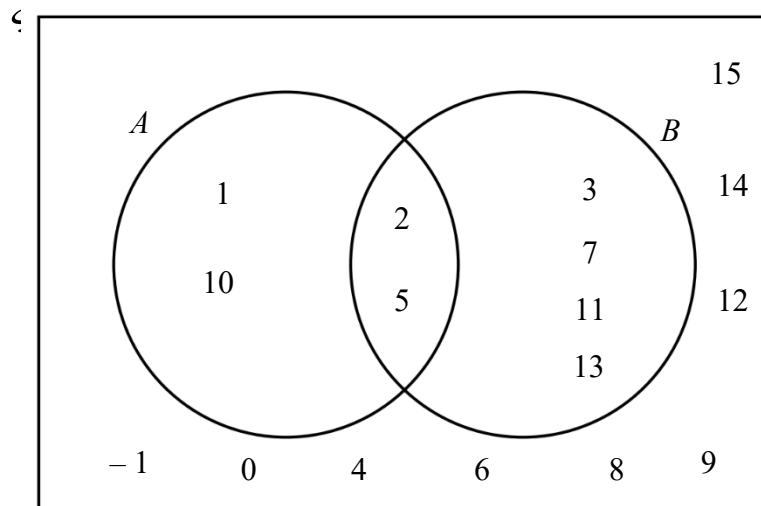
Answer g [2]

- 13 (a) Use set notation to describe the shaded region.



Answer [1]

- (b) The Venn diagram shows the universal set and the elements in each of its subsets.



- (i) Describe the elements of set A .

Answer

.....

 [1]

- (ii) List the elements in $A' \cap B'$.

Answer $A' \cap B' = \dots\dots\dots$ [1]

- (iii) Find $n\left[(A \cap B') \cup (A' \cap B)\right]$.

Answer $n\left[(A \cap B') \cup (A' \cap B)\right] = \dots\dots\dots$ [1]

- 14 (a) Solve the inequality $\frac{x+2}{4} \leq 5-2x < 13$.

Answer [3]

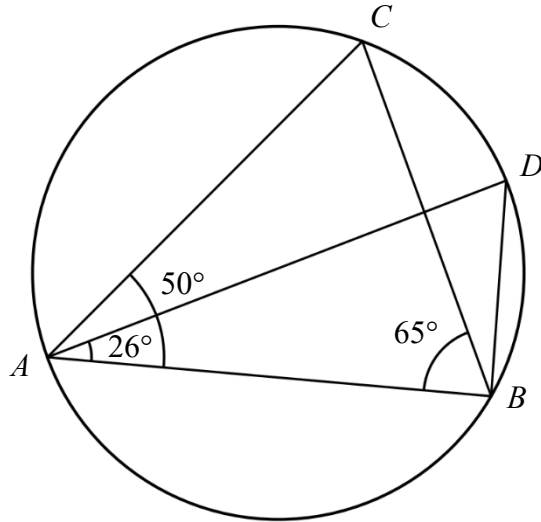
- (b) Represent your answer on the number line below.

Answer



[1]

- 15 A, B, C and D lie on a circle, with C and D on the same side of chord AB .
Angle $CAB = 50^\circ$, angle $ABC = 65^\circ$ and angle $DAB = 26^\circ$.



- (a) Find angle ADB .
Give a reason for each step of your working.

Answer angle $ADB = \dots\dots\dots$ [2]

- (b) Explain why AD does not pass through the centre of the circle.

Answer

.....

 [2]

16 5 cm on a map represents 2 km.

(a) Write the scale of the map in the form 1 : n .

Answer 1 : [1]

(b) An actual river is 12 km long.

Find the length, in centimetres, representing it on the map.

Answer cm [1]

(c) The area of a forest measures 8 cm² on the map.

Find the actual area of the forest, in square kilometres.

Answer km² [2]

BLANK PAGE

NAME:

NO:

CLASS:

RIVERSIDE SECONDARY SCHOOL**PRELIMINARY EXAMINATION 2026****SUBJECT : MATHEMATICS****SUBJECT CODE /PAPER : 4052/01****LEVEL/STREAM : Sec 4 Express, 5 Normal (Academic)****Part B****Answer all questions.**

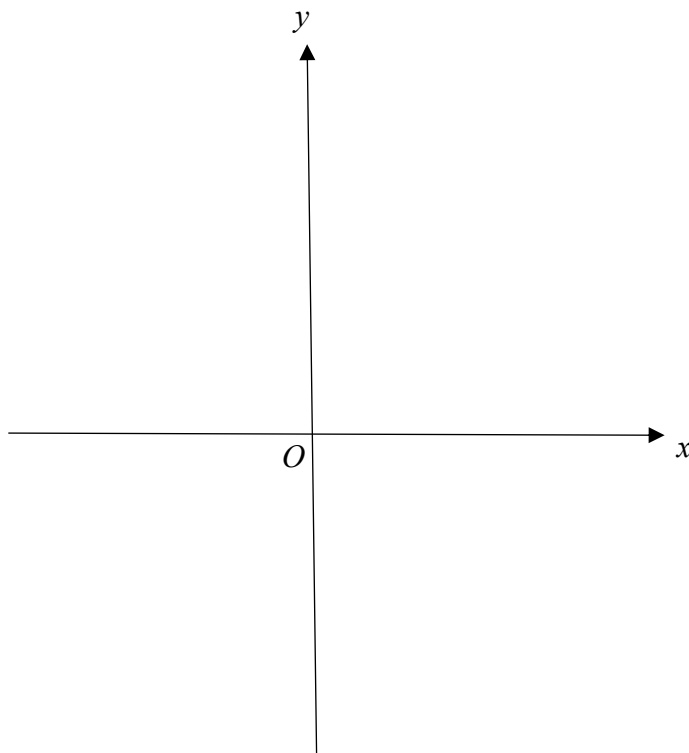
17 $y = -x^2 + 8x + 3$

- (a) Express y in the form $a(x-h)^2 + k$, where a , h and k are constants.

Answer $y = \dots\dots\dots$ [2]

- (b) Hence, sketch the graph of $y = -x^2 + 8x + 3$, showing the coordinates of the turning point and the y -intercept.

Answer



[2]

- 18 The first four terms of a sequence of fractions are $\frac{2}{3}$, $\frac{5}{8}$, $\frac{8}{13}$ and $\frac{11}{18}$.

(a) Write down the next term in the sequence.

Answer..... [1]

(b) Find, in terms of n , an expression for the n th term, T_n .

Answer $T_n =$ [1]

(c) Determine, showing your working, whether $\frac{503}{800}$ can be a term of this sequence.

Answer

[2]

19 Factorise completely

(a) $6x^2 + x - 12,$

Answer [2]

(b) $6x^2 - 9x + 4xy - 6y.$

Answer [2]

20 Mr Tan invests \$12 000 at a rate of 8% per annum, interest compounded quarterly.

(a) Explain why the interest earned after 1 year is not equal to 8% of \$12 000.

Answer

.....

 [1]

(b) Calculate the amount of interest Mr Tan will receive from the investment at the end of 4 years.

Answer \$ [3]

- 21** The list shows the number of laps completed by a swimmer over 8 training sessions, given in ascending order:

$$p \quad 2p \quad 16 \quad 19 \quad 23 \quad 27 \quad 30 \quad \frac{q}{2}$$

The mean number of laps is 21.

The range of the data is 29.

Form two equations in p and q , and hence find the value of p and of q .

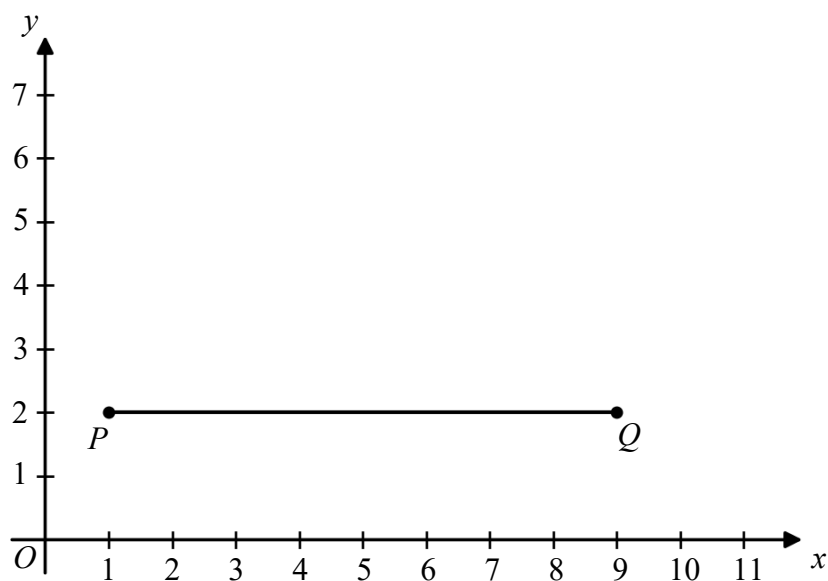
Answer

Answer $p = \dots\dots\dots, q = \dots\dots\dots$ [4]

- 22 The coordinates of P are $(1, 2)$ and Q are $(9, 2)$.

A line passes through Q with gradient $\frac{5}{2}$.

This line intersects the line $x = 11$ at point R .



- (a) Find the coordinates of R .

Answer (.....,) [2]

- (b) P , Q and R are three vertices of parallelogram $PQRS$.
Find the coordinates of S .

Answer (.....,) [1]

- (c) Calculate the area of parallelogram $PQRS$.

Answer units² [1]

- 23 (a) (i) Write 216 as a product of its prime factors.

Answer $216 = \dots\dots\dots$ [1]

- (ii) The highest common factor of two numbers is 12.
The lowest common multiple of the two numbers is 216.
Both numbers are greater than 15.
Find the two numbers.

Answer $\dots\dots\dots$ and $\dots\dots\dots$ [2]

- (b) Written as a product of its prime factors, $1728 = 2^6 \times 3^3$.

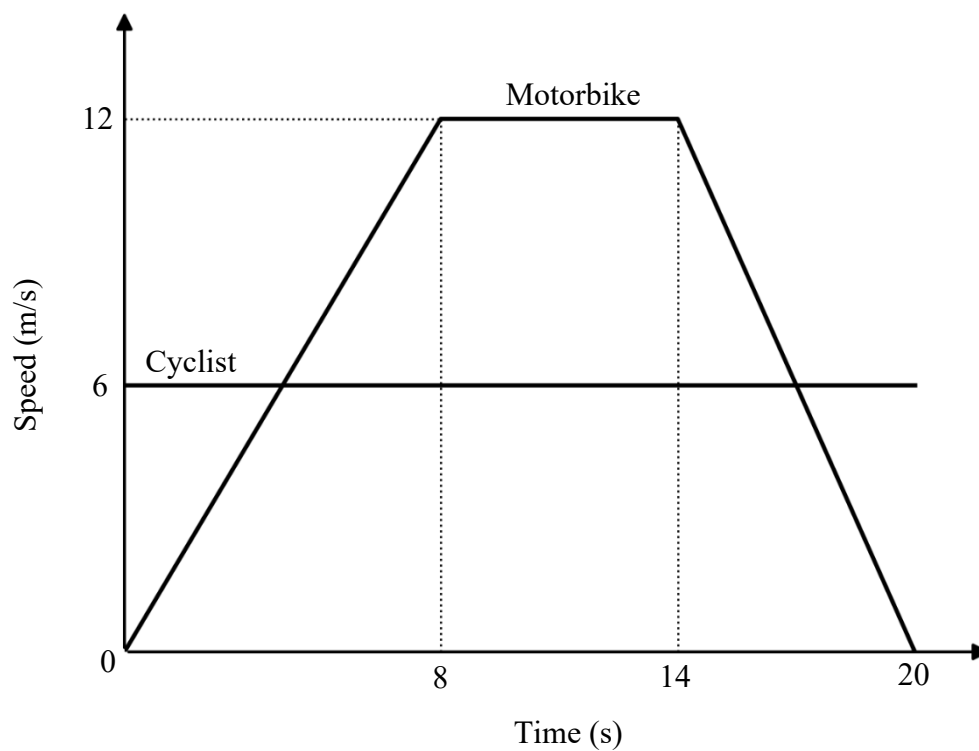
The number $1728 \div \frac{p^2}{q}$ is a perfect square, where p and q are

prime numbers, $p < q$.

Find the value of p and of q .

Answer $p = \dots\dots\dots$, $q = \dots\dots\dots$ [2]

- 24 The diagram shows speed-time graphs for a motorbike and a cyclist, travelling along the same road in the same direction.



- (a) Find the acceleration of the motorbike during the first 6 seconds.

Answer m/s^2 [1]

- (b) Without further calculation, state and explain whether the magnitude of the deceleration in the last stage is greater than, less than, or equal to the acceleration in the first stage.

Answer

The magnitude of the deceleration is greater than / less than / equal to the acceleration in the first stage because

.....

.....

..... [1]

- (c) When $t = 0$, the motorbike is 12 m behind the cyclist.

When $t = T$, the motorbike passes the cyclist.

Find the value of T .

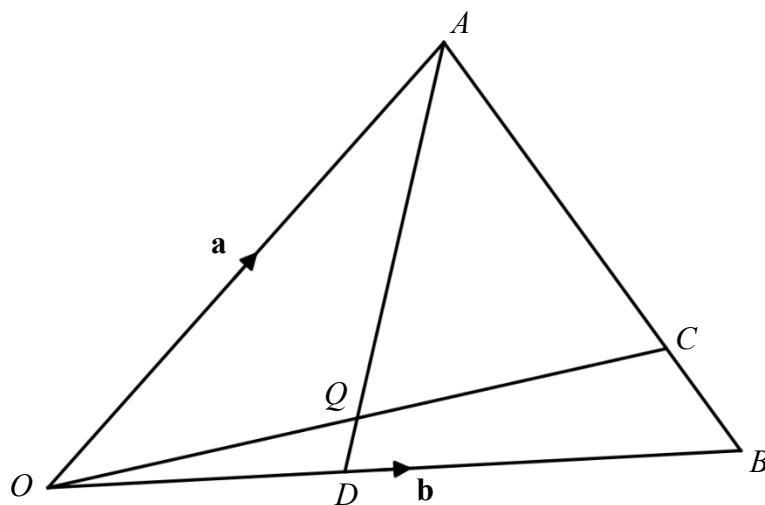
Answer $T =$ [3]

25 OAB is a triangle.

$\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$.

C is the point on AB such that $AC : CB = 3 : 1$.

D is the point on OB such that $OD : DB = 3 : 4$.



(a) Find, in terms of $\mathbf{a}$ and $\mathbf{b}$,

(i) $\overrightarrow{AB}$,

Answer $\overrightarrow{AB} = \dots\dots\dots$ [1]

(ii) $\overrightarrow{OC}$,

Answer $\overrightarrow{OC} = \dots\dots\dots$ [2]

(iii) $\overrightarrow{OD}$.

Answer $\overrightarrow{OD} = \dots\dots\dots$ [1]

(b) Q is a point on OC such that $\overrightarrow{OQ} = \frac{1}{8}\mathbf{a} + \frac{3}{8}\mathbf{b}$.

(i) Show that A , Q and D are collinear. [3]

(ii) Find the ratio of the area of triangle OAQ to the area of triangle ODQ .

Answer $\dots\dots\dots : \dots\dots\dots$ [1]

BLANK PAGE